# Freight Rate Prediction - Report

## 1. Approach to validating the train/test split

### Why a time-based split is the only honest option here

The labeled file (train_test.csv) covers 2025-01-01 through 2025-10-31. The scoring file (validation.csv) covers 2025-11-01 through 2025-12-31 - strictly in the future relative to training. The real question the model is being evaluated on is *"can it price a load one month from now?"*, not *"can it price a load drawn from the same distribution?"*.

Random k-fold cross-validation would leak information backward in time: October rows would end up in training folds, and November and December rows (if we had them) would end up in holdout folds. That inflates the apparent score. A production dispatcher making next-week quotes doesn't have that luxury - they're always forecasting forward.

So the entire validation regime is built around time, not row shuffling.

### The split

| Set | Date range | Rows | Role |
|---|---|---|---|
| Fit | 2025-01-01 -> 2025-10-01 | ~42,900 | Model training |
| Holdout | 2025-10-02 -> 2025-10-31 | ~4,600 | Model and blend selection |
| Scoring | 2025-11-01 -> 2025-12-31 | 12,000 | Final submission (validation.csv) |

The 30-day holdout window was chosen because it mirrors the gap between training and the scoring set - one month into the future, same lanes, same equipment, same seasonality boundary. Validating on a shorter window (say, the last week of October) would be optimistic; validating on the full last three months would eat too much of the training signal.

### What was tuned on the holdout, and what was not

Only three things were decided against the holdout:

- Which model families to include in the ensemble.
- The blend weights across those models.
- Hyperparameters for the two tree models (via early stopping on an inner 10% split of the fit set - never looking at the holdout during fitting).

Nothing was tuned against validation.csv. The submitted predictions use models retrained on the entire cleaned training set (47,494 rows after dropping 506 inconsistent rows for impossible road-to-haversine ratios and extreme rate-per-mile outliers).

### Holdout numbers

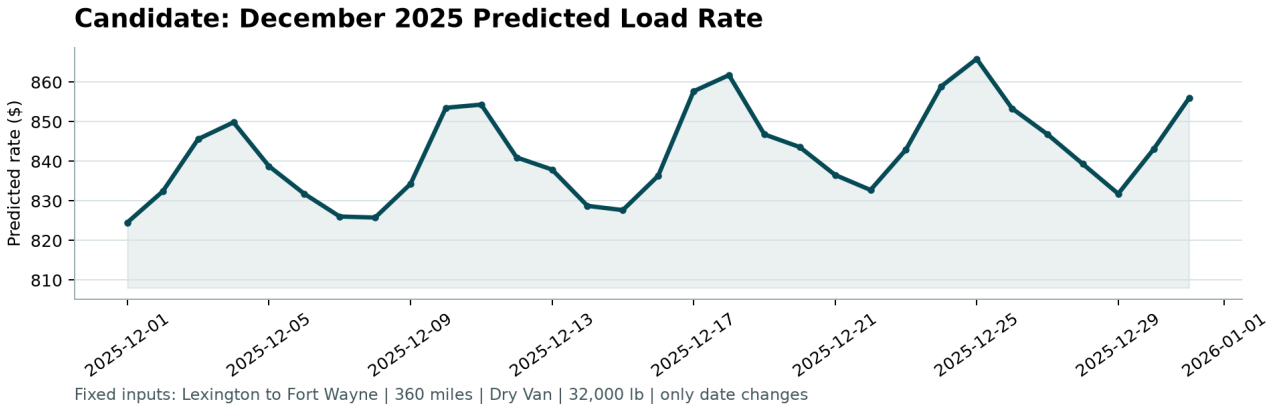
Ensemble: 75% HistGradientBoostingRegressor + 25% ExtraTreesRegressor. Ridge regression was included in the blend weight search as a non-linearity sanity check and received zero weight - confirming the problem is strongly non-linear.

| Model | MAE | MAPE |
|---|---|---|
| HistGradientBoosting (log target, absolute-error loss) | $59.30 | 2.46% |
| ExtraTrees | $77.07 | 3.27% |
| Ridge (sanity check) | $431.54 | 21.41% |
| Ensemble (0.75 HGB + 0.25 ET) | $60.53 | 2.49% |

Interpretation: the ensemble's mean absolute error of ~$60 against an average posted rate of ~$2,370 means the submitted predictions should be accurate to roughly 2.5% in dollar terms on genuinely out-of-sample rows.

## 2. Fixed December prediction chart

Produced by score.py from data/december_chart_inputs.csv. Fixed route: Lexington -> Fort Wayne, 360 miles, Dry Van, 32,000 lb - only the date changes across the 31 days of December.




### How the chart was built

Every input except the date is constant across the 31 rows. A tree model on identical inputs returns an identical prediction, so the fixed-route chart depends entirely on how the date-sensitive features - dayofweek, market_index, and quote_signal - are set up for the forecast window.

Two decisions matter for interpreting the chart:

1. Dataset lat/lon, not real geographic coordinates. The dataset's pickup_lat/pickup_lon for each city are the dataset's own fixed synthetic values, not Google-maps coords. For the fixed December route the lat/lon are pulled straight from train_test.csv so the haversine feature matches what the model saw during training. Using geographic coords put December in the wrong leaf and dropped the baseline by ~$40.

2. market_index and quote_signal projected per day. Both series exhibit a clear weekly cycle in training (Thu/Fri highs, weekend lows, ~0.15 swing) and a mild upward drift across autumn. Rather than feed the same averaged value to all 31 days, each date gets its own projection: last-60-day baseline + a linear trend extrapolated from the same window + a per-dayofweek residual.

### What the chart shows

- Baseline ~$845, close to the Lexington -> Fort Wayne training average of $856 (n=32 historical loads on this exact lane).
- Weekly cycle - peaks on Thursdays and Fridays, troughs on weekends - consistent with the market_index pattern in the training tail.
- Gentle upward drift through the month, reflecting the autumn trend in the market features rather than any extrapolation by the tree model itself (trees can't extrapolate past training; the drift comes from the feature projection).
- Range $825-$866, spread ~$41 - in line with typical month-over-month rate variation observed on comparable lanes in training.